

Supplementary Information: Aftershock deficiency of induced earthquake sequences during rapid mitigation efforts in Oklahoma

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Abstract

In this supplementary material, we briefly discuss the statistical analysis of earthquake clustering and how it is used to separate clustered from background events (Fig. S1-S4). We then show injection rates and earthquake activity for regions with active mitigation and deficient aftershock productivity, including the *M*5.8 Pawnee, *M*4.7 Cherokee and *M*5.7 Prague events as well as a map of mainshocks with and without active mitigation (Fig. S5-S6). Figure S7 to S11 summarize results from the benchmark poroelastic model as well as the sensitivity study for the layered poroelastic model described in the main manuscript. Lastly, we show well locations and seismicity during three different time windows before, during and after the *M*5 Cushing event. A more complete picture of the unfolding Cushing aftershocks and near-by injection activity is provided through an animation available online (animation S1).

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S1. Sensitivity of clustering analysis and aftershock identification

To separate the observed seismicity into background and clustered events, we use the distribution of nearest-neighbor event pairs and compare the observed clustering characteristics with expectations from random poissonian earthquakes (Fig. S1) (*Zaliapin et al.*, 2008; *Zaliapin and Ben-Zion*, 2013). Commonly, space-time distances between event pairs can be described by a 2D bi-modal distribution. The first mode at small interevent times and distances highlights clustered events (e.g. aftershocks), whereas the second mode at larger distances is comprised of background events. Fig. S1 C & E highlight that the observed background mode for the California catalog corresponds to expectation from a randomized catalog. We use the 99th percentile of nearest-neighbor distances from the randomized catalogs to separate background and clustered events, which allows for a clear separation between the two modes in California and for the relocated catalog in Oklahoma Fig. S1C&F.

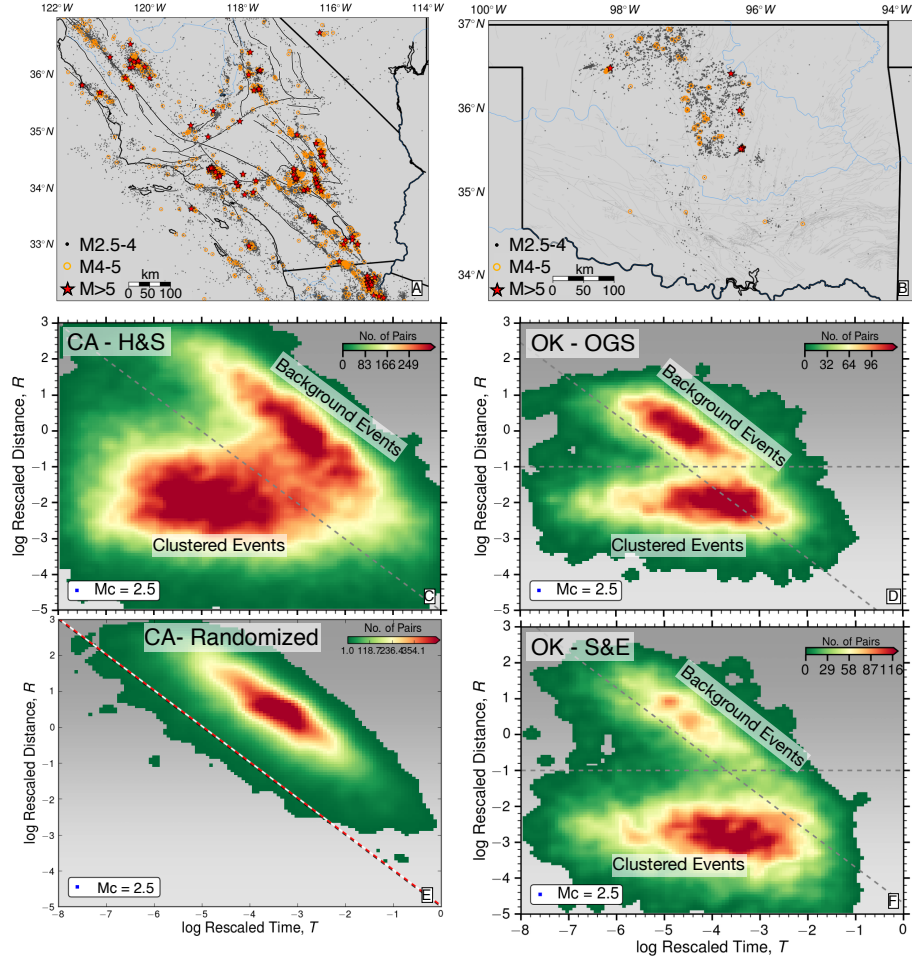


Figure S1: Maps of seismicity in California (A) and Oklahoma (B) and rescaled distance-time plots for the identification of clustered events (C-F). We use a relocated catalog for Southern California (CA-H&S) from *Hauksson et al.* (2012) and both a standard (OK-OGS) and relocated catalog (OK-S&E) for Oklahoma, the latter from *Schoenball and Ellsworth* (2017). The colors in C-F correspond to the number of event pairs at a specific space-time distance. Background and clustered events are separated by gray dashed lines with clustered events occurring at small space-time distances. Note that the space-time event pair distribution for the standard OGS catalog cannot be separated by a simple diagonal line which is likely a result of high location uncertainty and varying background rates. For this catalog, we also test fixed distance windows which are a function of mainshock magnitude (see Fig. S2).

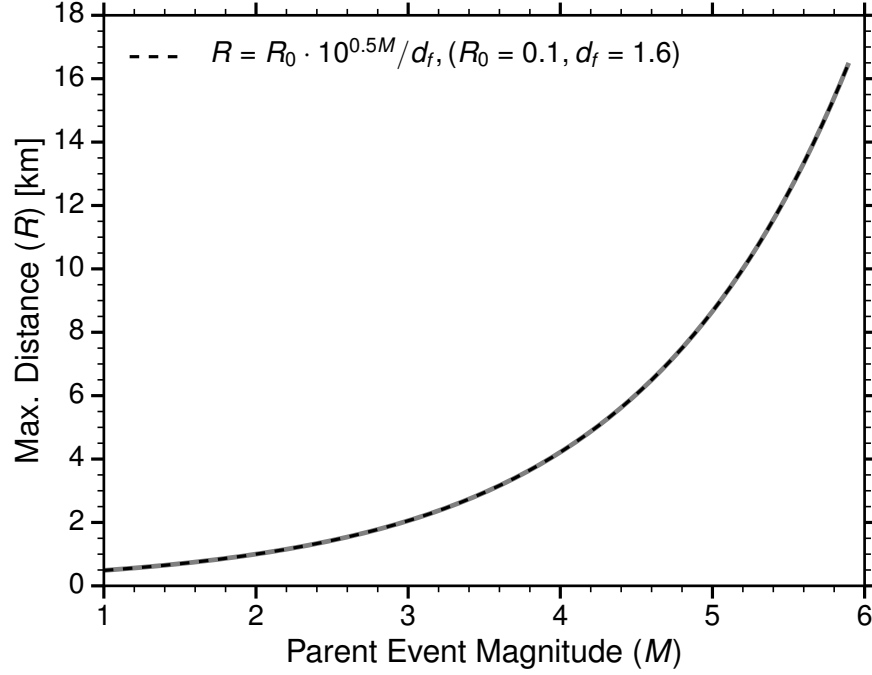


Figure S2: Maximum spatial window for aftershock detection as a function of parent event magnitudes. This window is only used for a sensitivity analysis of the described separation into clustered and background events.

For the standard earthquake catalog in Oklahoma, on the other hand, the two observed modes of rescaled space-time distances cannot be as easily separated. For this catalog, we test (in addition to the separation based on expectation from randomized catalogs) the influence of a maximum distance separation (Fig. S2) which is more in line with classical, window-based clustering analyses (e.g. *Gardner and Knopoff*, 1974). We find that the aftershock statistics and the general trend of deficient aftershock productivity for larger magnitude mainshocks remains robust for both catalogs and the different analyses steps in Oklahoma.

We test the influence of strongly varying background rates in Oklahoma on the clustering analysis by separating the catalog into three year time intervals

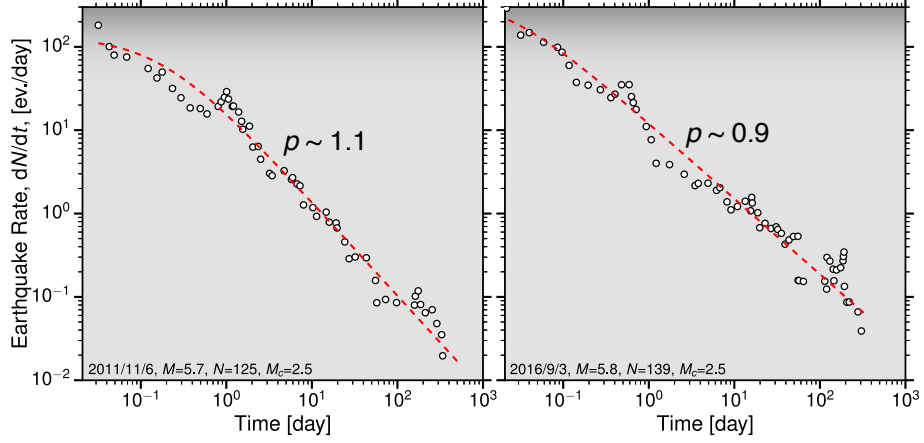


Figure S3: Two examples of aftershock rates and Omori-fits after the M5.7 Prague event (left) and the M5.8 Pawnee event. The Omori parameters were determined by using a maximum likelihood method (*Ogata*, 1999). Temporal decay exponents, p , are shown in each plot and correspond to the slope of the red dashed lines.

from 2010 to 2012, 2013 to 2015 and 2016 to 2018 (Fig. S4). The clustering analysis was performed on each sub-catalog and mainshocks and corresponding number of aftershocks were combined at the end to examine the average trends. The overall productivity trend and detection of mainshocks with deficient aftershocks are consistent between the entire catalog and sub-catalogs. We tested both two and three year time windows and found comparable results. This consistency highlights the robustness of the applied aftershock detection method even when background rates vary strongly as is the case in Oklahoma.

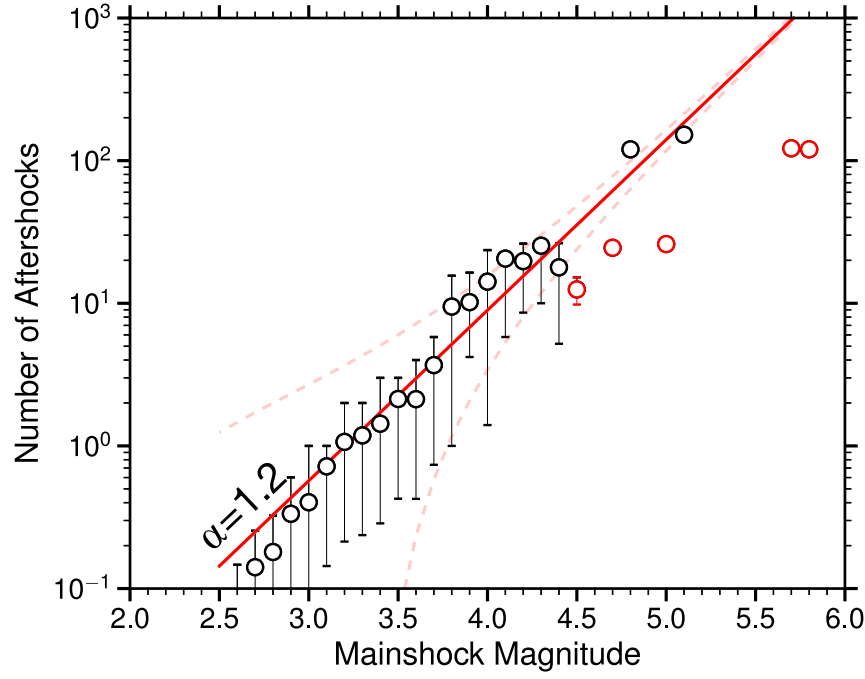


Figure S4: Aftershock productivity in Oklahoma for mainshock magnitudes between 2.5 and 5.8, with exponential fit (red line) and confidence interval (dashed line) and mainshocks with deficient aftershock activity, highlighted by red markers. This figure is analogous to Figure 1B in the main text but the clustering analysis was performed in 3 year time intervals from 2010-2012, 2013-2015 and 2016-2018 to test the influence of varying background rates on the determined productivity values.

S2. Mainshock locations and injection rate variations

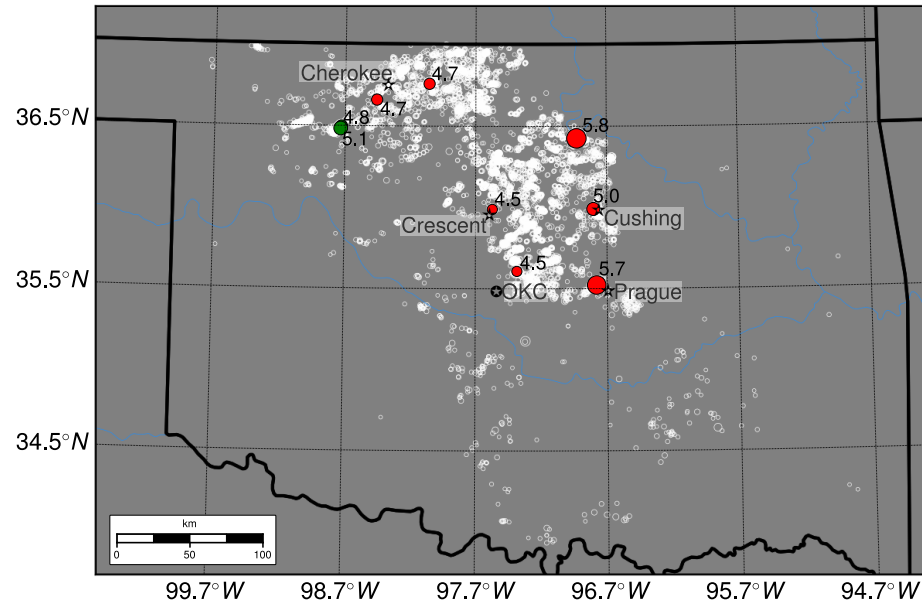


Figure S5: Map of mainshocks with and without active mitigation in red and green. The mainshocks with active mitigation also exhibit low productivity aftershock sequences.

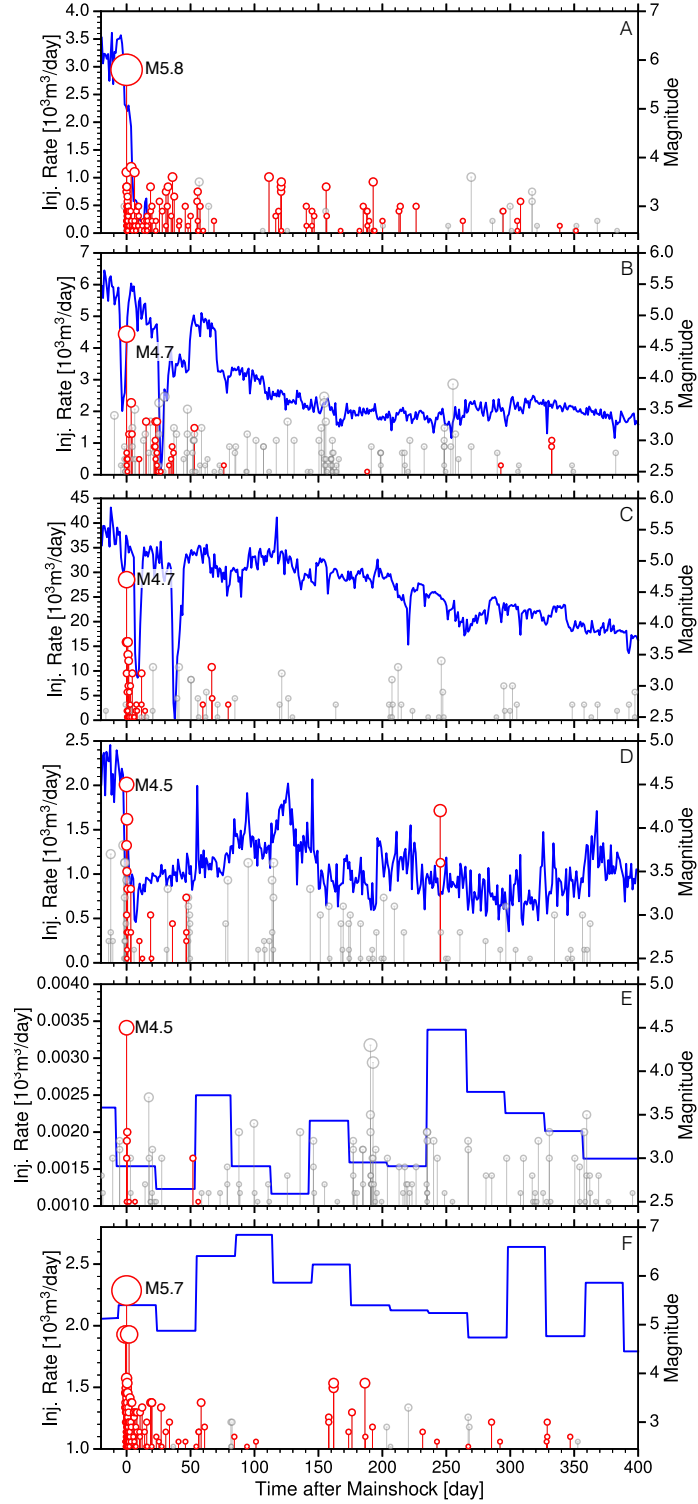


Figure S6: Injection rates (blue) and aftershock magnitudes (red) for sequences with deficient aftershock numbers. The gray circles show background events within 15 km from each mainshock.

S3. Poroelastic stress changes and benchmark tests

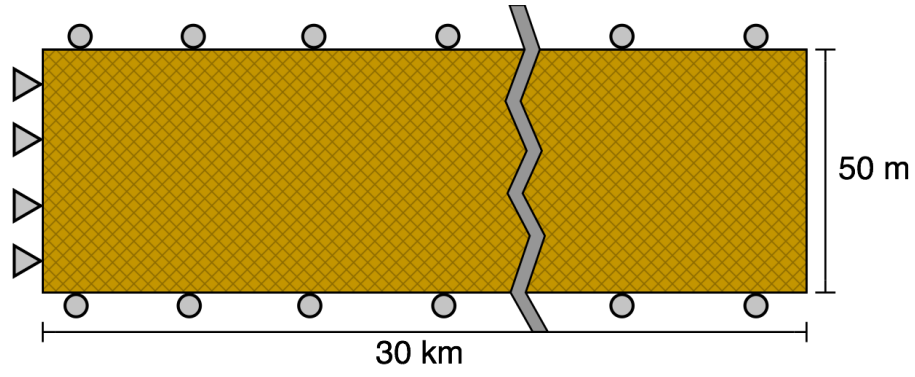


Figure S7: Geometry and boundary conditions for the simple benchmark model for which analytical solutions are available from *Helm* (1994) and *Rudnicki* (1986).

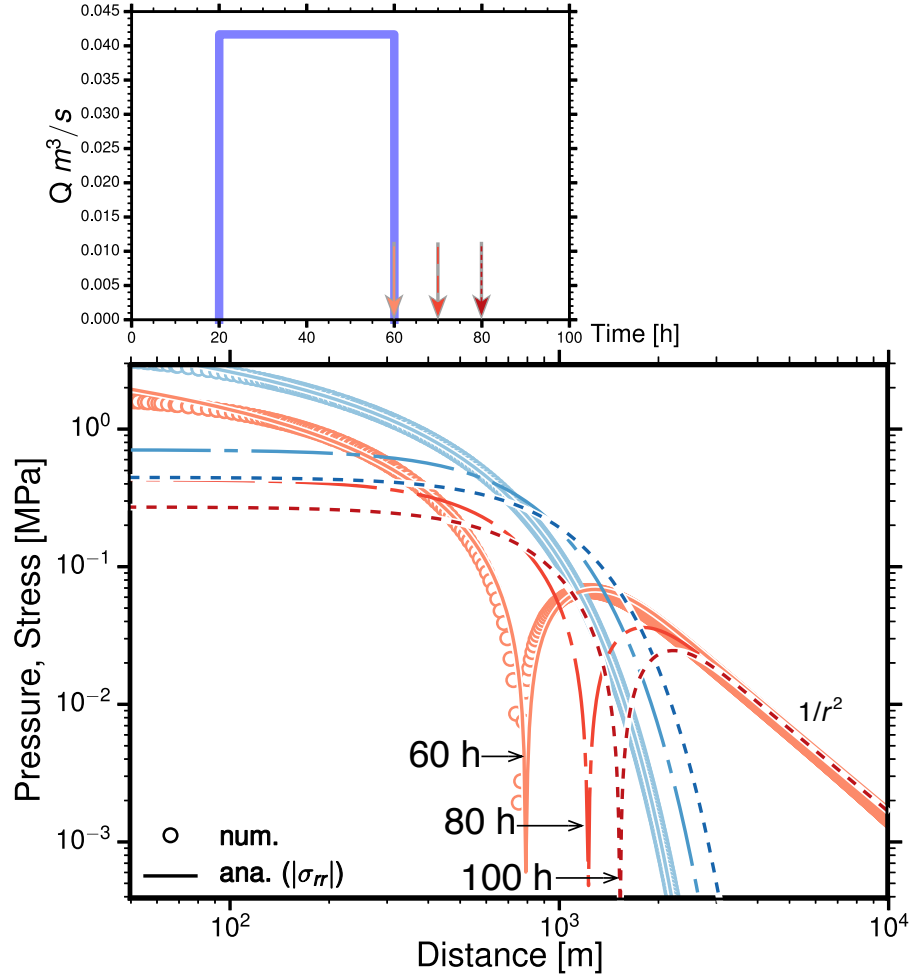


Figure S8: Injection rate function (top) and spatial decay of fluid pressures (blue) and solid stresses in the radial direction (red) within the benchmark model. Analytical solutions are shown by solid and dashed lines and numerical solutions are highlighted by blue and red circles. Injection is stopped after 60 h. After that, pore-pressure decreases rapidly within 700 m from the well and poroelastic stresses decrease out to more than 2 km.

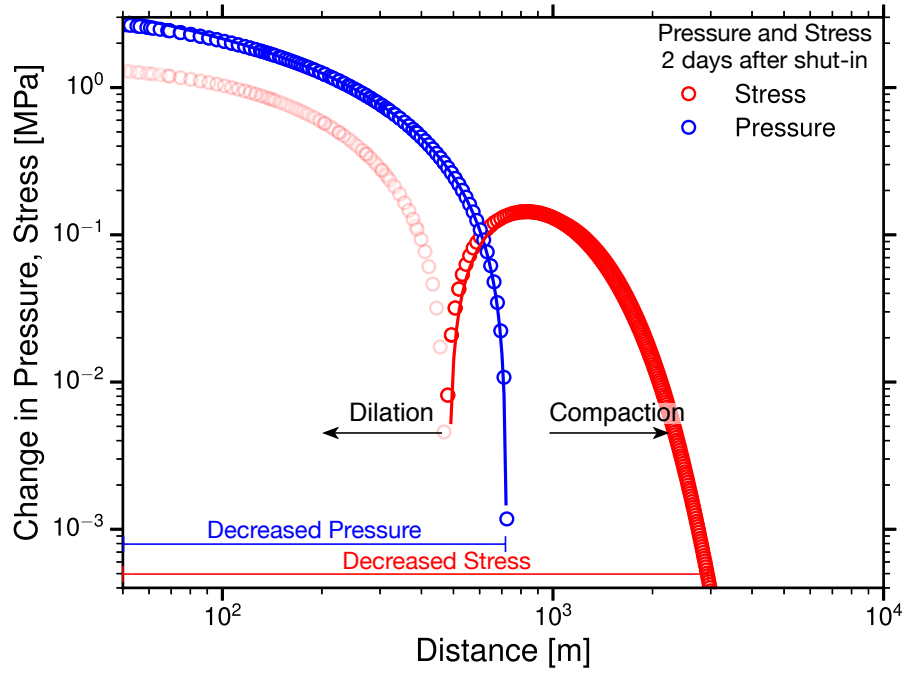


Figure S9: Distance extent of fluid pressure (blue) and solid stress (red) decrease in the benchmark model 2 days after injection stop. Numerical solutions are shown by open circles and analytical solutions by solid lines. The solution from this benchmark model could be considered an upper bound for expected poroelastic stress changes within the injection reservoir after well shut-in.

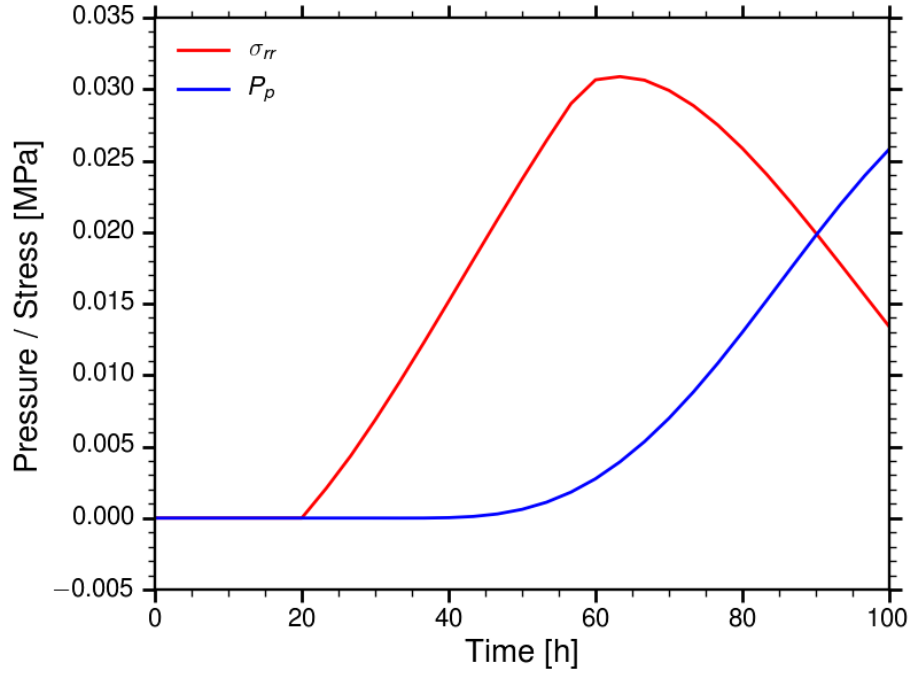


Figure S10: Example of reduction in solid stresses in the basement at 4 km distance from the well after injection was stopped. The solution is based on the layered poroelastic model described in the main manuscript. Note that fluid pressures continue to increase at this distance from the well because of the time delay of the diffusive process.

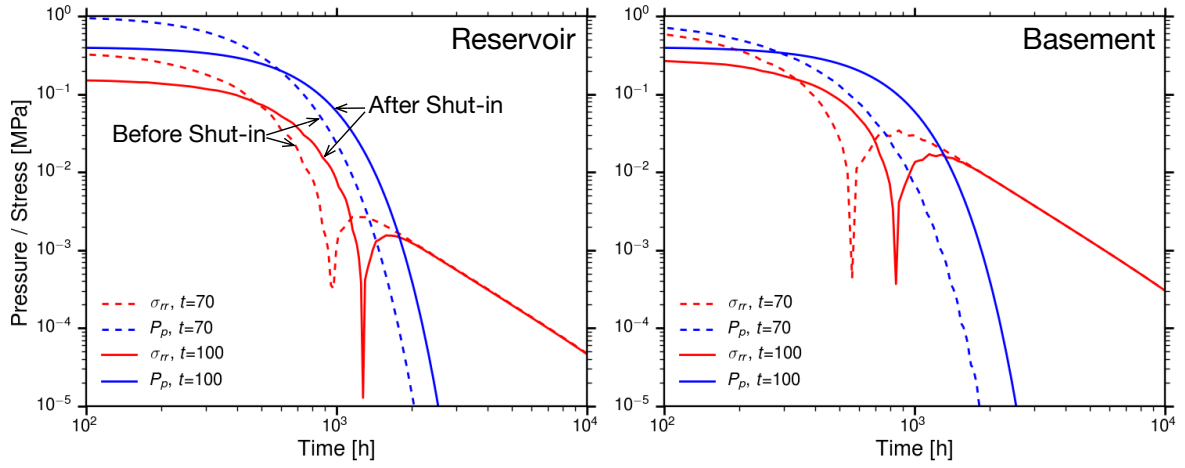


Figure S11: Spatial changes in fluid pressure (blue) and solid stress (red) within reservoir and basement before (dashed) and after (solid) well shut-in. The solution is based on the layered poroelastic model described in the main manuscript but with higher coupling coefficient within the basement. Note that the relatively higher elastic stresses in the basement are due to the increase in elastic moduli assuming that poroelastic coupling remains constant (bulk modulus/matrix modulus = constant).

S4. Seismicity and injection well locations near Cushing, Oklahoma in 2016

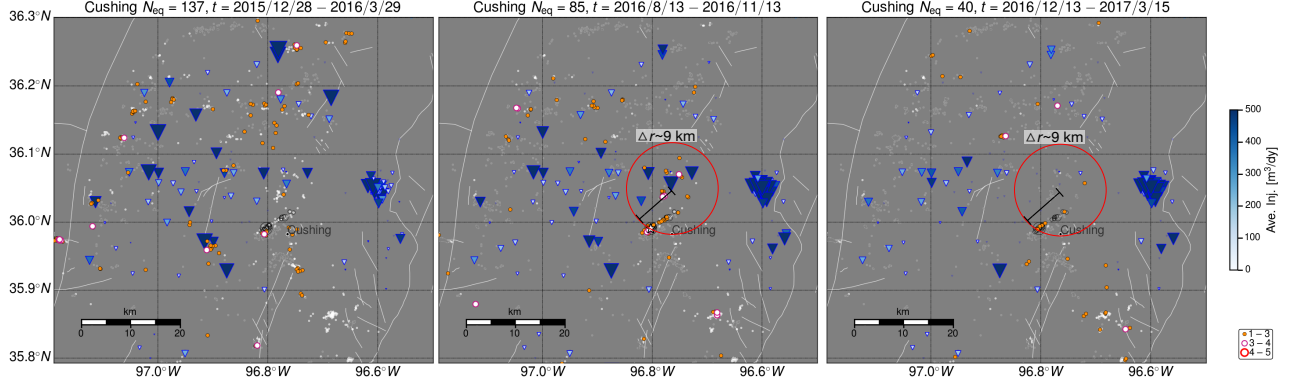


Figure S12: Seismicity and well locations within the Cushing area before (left) during (center) and after (right) the $M5$ Cushing earthquake sequence. Injection wells are colored according to average monthly injection rates (see color bar). Orange and red circles highlight earthquake magnitudes (see legend), the mainshock is highlighted by a white star. The Cushing events were closely connected to an increase in injection in wells about 9 km north of the mainshock. Mitigation in form of shut-in and injection rate decrease in these wells also led to a rapid decrease in seismicity rates after the mainshock.

S5. References

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